ESTRUTURAS DE CONCRETO ARMADO II

Aula 01b — 20 de agosto de 2026

Exercício

1. Considerando a planta de forma estrutural apresentada, pede-se:

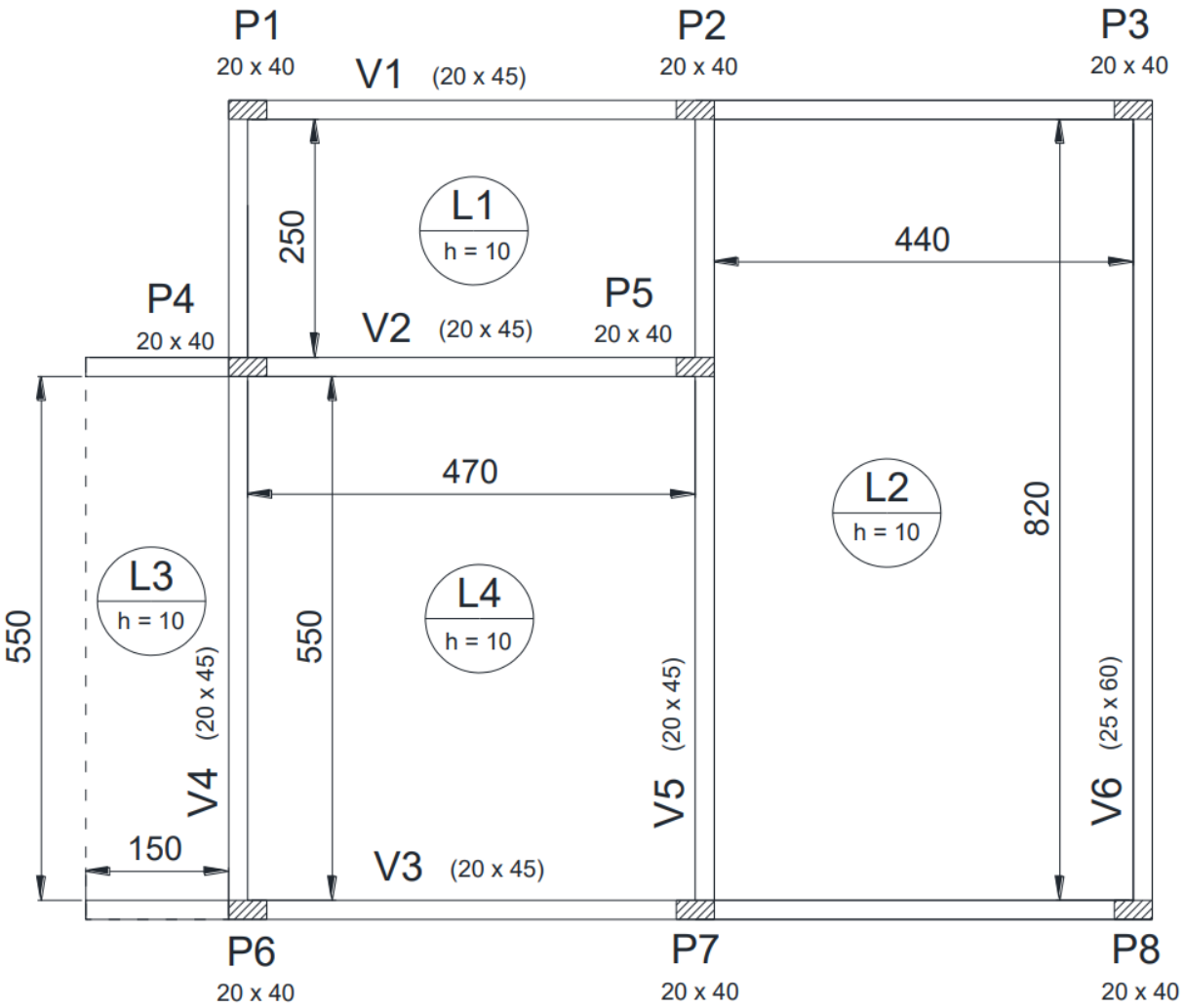


Figura 1 – Forma pavimento tipo.

- classificar as lajes em armadura me uma ou duas direções;
- indicar a vinculação de cada uma das lajes;
- considerando carregamento permanente $g = 3,5\ kN/m^2$ (peso próprio da laje + peso próprio do piso e revestimentos) e carregamento variável normal $q = 3\ kN/m^2$ (edifícios comerciais), calcular as reações de apoio das lajes.

- considerando carregamento permanente $g = 3,5\ kN/m^2$ e carregamento variável normal $q = 3\ kN/m^2$, calcular e compatibilizar os momentos atuando nas lajes.

1. Determinação dos vãos efetivos e classificação

l_x menor dimensão

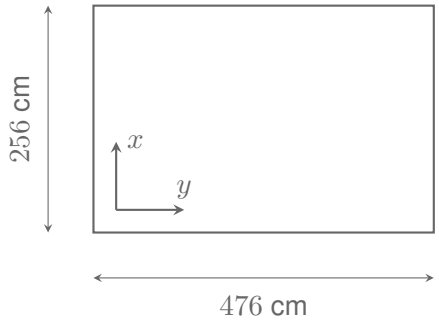
l_x maior dimensão

b_{w_0} zero é a indicação da viga

$\lambda \leq 2$ armada em duas direções

$\lambda > 2$ armada em uma direção

L1



a) Direção horizontal

$$l_0 = 470\ cm$$

$$a_1 \leq \begin{cases} \dfrac{b_{w_4}}{2} = \dfrac{20}{2} = 10\ cm \\ 0,3h = 0,3 \times 10 = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_5}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 476\ cm$$

b) Direção vertical

$$l_0 = 000\ cm$$

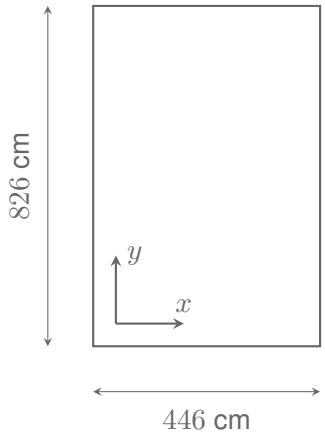
$$a_1 \leq \begin{cases} \dfrac{b_{w_1}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_2}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 256\ cm$$

$$\lambda = \frac{l_y}{l_x} = \frac{476}{256} \approx 1,86 \leq 2 \rightarrow \text{armada em duas direções}$$

L2



a) Direção horizontal

$l_0 = 440\ cm$

$$a_1 \leq \begin{cases} \frac{b_{w_1}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \frac{b_{w_3}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$l_{ef} = l_0 + a_1 + a_2 = 446\ cm$

b) Direção vertical

$l_0 = 820\ cm$

$$a_1 \leq \begin{cases} \frac{b_{w_5}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

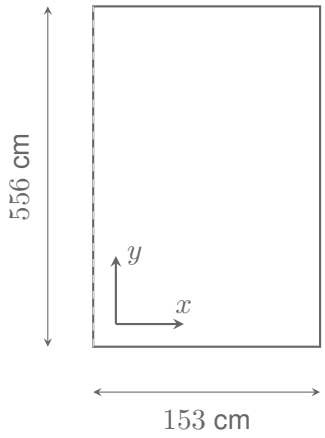
$$a_2 \leq \begin{cases} \frac{b_{w_6}}{2} = \frac{25}{2} = 12,5\ cm \\ 0,3h = 3\ cm \end{cases}$$

$l_{ef} = l_0 + a_1 + a_2 = 826\ cm$

$\lambda = \frac{l_y}{l_x} = \frac{826}{446} \approx 1,85 \leq 2 \rightarrow$ armada em duas direções

Continuação - para casa

L3



a) Direção horizontal

$$a_1 \leq \begin{cases} \frac{b_{w_4}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$l_0 = 150\ cm$

$$l_{ef} = l_0 + a_1 + a_2 = 153\ cm$$

b) Direção vertical

$$l_0 = 550\ cm$$

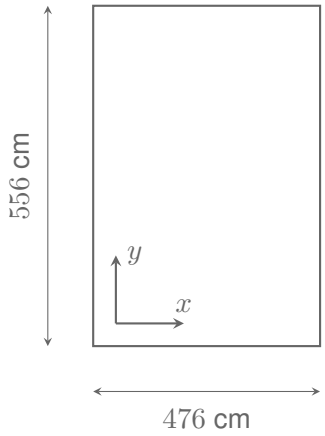
$$a_1 \leq \begin{cases} \dfrac{b_{w_2}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_3}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 556\ cm$$

$$\lambda = \frac{l_y}{l_x} = \frac{556}{153} \approx 3,63 > 2 \rightarrow \text{armada em uma direção}$$

L4



a) Direção horizontal

$$l_0 = 470\ cm$$

$$a_1 \leq \begin{cases} \dfrac{b_{w_2}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_3}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$l_{ef} = l_0 + a_1 + a_2 = 476\ cm$$

b) Direção vertical

$$l_0 = 550\ cm$$

$$a_1 \leq \begin{cases} \dfrac{b_{w_4}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

$$a_2 \leq \begin{cases} \dfrac{b_{w_5}}{2} = 10\ cm \\ 0,3h = 3\ cm \end{cases}$$

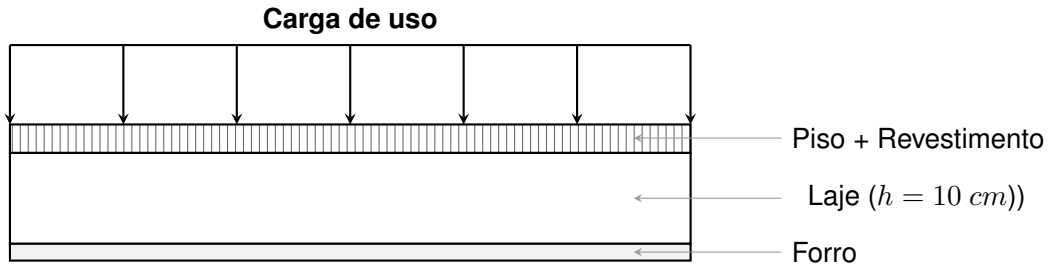
$$l_{ef} = l_0 + a_1 + a_2 = 556\ cm$$

$$\lambda = \frac{l_y}{l_x} = \frac{556}{476} \approx 1,17 \leq 2 \rightarrow \text{armada em duas direções}$$

Aula 02 — 26 de agosto de 2026

2. Determinação do carregamento nas lajes

   É preciso calcular o carregamento permanente para cada laje, mas como h = 10 cm para todos não precisou repetir.



   $g_{PP} = \gamma_{ap} \cdot h = 25 \times 10 \cdot 10^{-2} = 2,5\ kN/m^2 \rightarrow \gamma_{ap} = 25\ kN/m^3$ (ABNT NBR 6120, 2019, tabela 1)

   $g_{P+R} = 1\ kN/m^2$ (ABNT NBR 6120, 2019, tabela 4)

   $g_{forro} = 0,25\ kN/m^2$ (ABNT NBR 6120, 2019, tabela 8)

   $g = g_{PP} + g_{P+R} + g_{forro} = 2,5 + 1 + 0,25 = 3,75\ kN/m^2$

   $q = 3\ kN/m^2$ (ABNT NBR 6120, 2019, tabela 10)

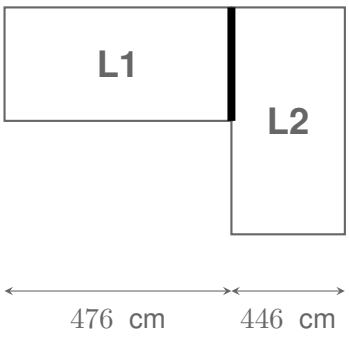
   $g + q = 3,75 + 3 = 6,75\ kN/m^2$

3. Vinculação das lajes

   Multiplica por três por causa da recomendação do livro IBRACON.

   L1

   a) Vizinhança V5

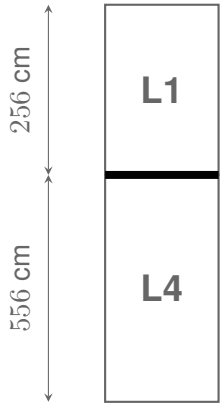


$\ell_1 < 3\ell_2$

$476 < 3 \times 446$

$476 < 1338 \rightarrow$ ambas engastadas

b) Vizinhança V2



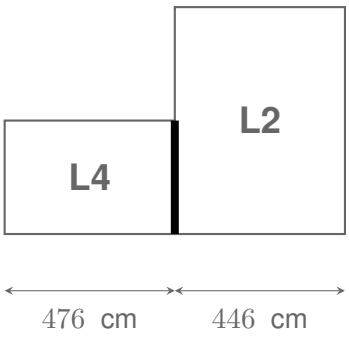
$\ell_1 < 3\ell_2$

$556 < 3 \times 256$

$556 < 768 \rightarrow$ ambas engastadas

L2

a) Vizinhança V5



$\ell_1 < 3\ell_2$

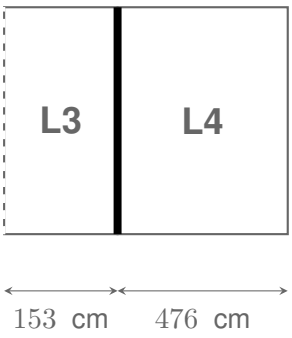
$476 < 3 \times 446$

$476 < 1138 \rightarrow$ ambas engastadas

L3

Por ser uma laje em balanço precisa de engaste.

a) Vizinhança V4



$\ell_1 < 3\ell_2$

$476 < 3 \times 153$

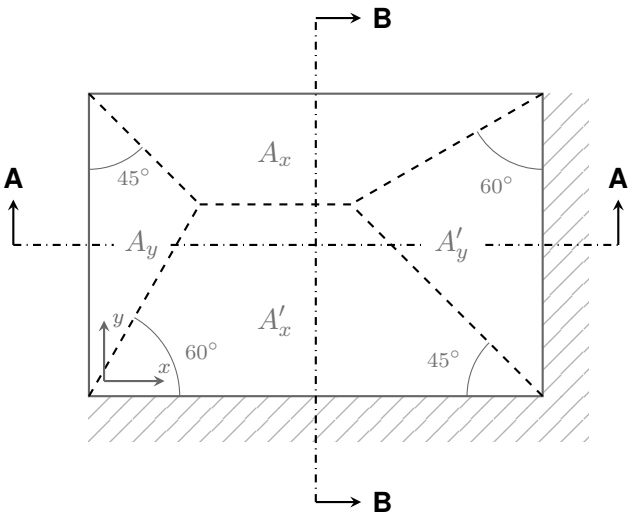
$476 < 459 \rightarrow$ ambas engastadas

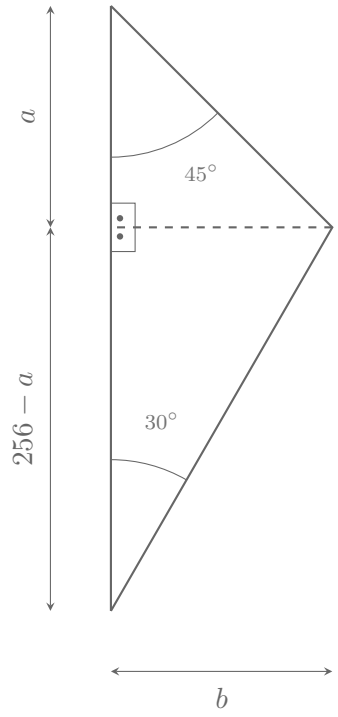
Maior laje fica apoiada e a menor fica engastada.

4. Reações verticais nas lajes

a) Charneiras plásticas

L1





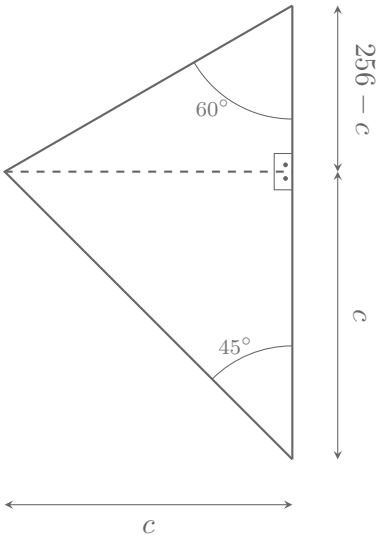
$$\cancel{\tan 45} = 1 \ \frac{b}{a} \longleftrightarrow b = a$$

$$\tan 30 = \frac{a}{256 - a}$$
$$\frac{\sqrt{3}}{3}(256 - a) = a$$
$$\frac{256\sqrt{3}}{3} = (1 + \frac{\sqrt{3}}{3})a$$
$$a \approx 93,70\ cm$$

$$L = 256 - a$$
$$L \approx 256 - 93,70$$
$$L \approx 162,30\ cm$$

$$A_y = \frac{bh}{2} \approx \frac{256 \times 93,7}{2} \approx 1,2\ m^2$$

$$V_y = \frac{(g+q)A_y}{\ell_x} \approx \frac{6,75 \times 1,2}{2,56} \approx 3,16\ kN/m$$

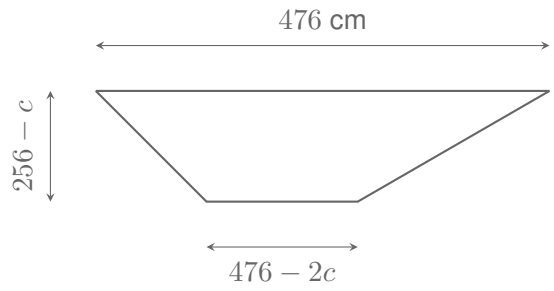


$$\tan 60 = \frac{c}{256 - c}$$
$$256\sqrt{3} = (1 + \sqrt{3})c$$
$$c \approx 162,30\ cm$$

$$L = 256 - c$$
$$\approx 256 - 162,30$$
$$\approx 93,70\ cm$$

$$A'_y = \frac{bh}{2} \approx \frac{256 \times 162,30}{2} \approx 2,08\ m^2$$

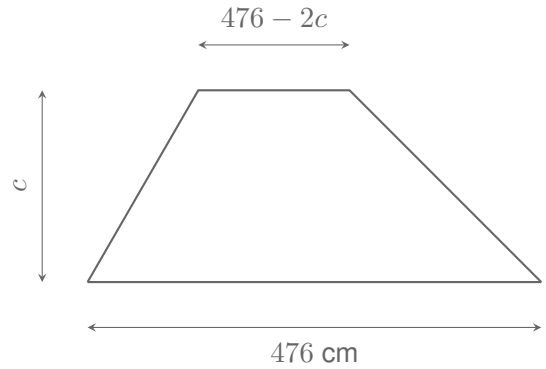
$$V'_y = \frac{(g+q)A'_y}{\ell_x} = \frac{6,75 \times 2,08}{2,56} \approx 5,48\ kN/m$$



$$b = 476 - 2c \approx 476 - 93,7 - 162,3 = 220\ cm$$

$$A_x = \frac{(B+b)h}{2} \approx \frac{(476+220)93,7}{2} \approx 3,26\ m^2$$

$$V_x = \frac{(g+q)A_x}{\ell_y} \approx \frac{6,75 \times 3,26}{4,76} \approx 4,62\ kN/m$$



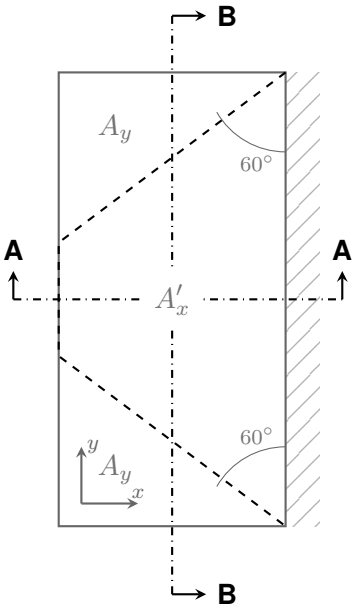
$$A'_x = \frac{(B+b)h}{2} \approx \frac{(476+220)162,3}{2} \approx 6,65\ m^2$$

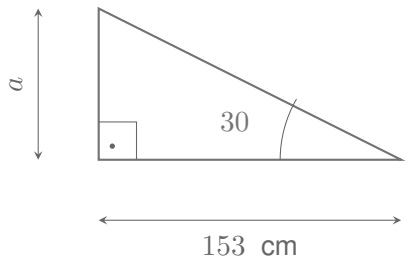
$$V'_x = \frac{(g+q)A_x}{\ell_y} \approx \frac{6,75 \times 5,65}{4,76} \approx 8,01\ kN/m$$

Continuação - para casa

L2

L3





$$\tan 30 = \frac{a}{1,53}$$
$$a \approx 0,88\ m$$

$$A_y = \frac{bh}{2} \approx \frac{1,53 \times 0,88}{2} \approx 0,68\ m^2$$

$$V_y = \frac{(g+q)A_y}{\ell_x} \approx \frac{6,75 \times 0,68}{1,53} \approx 2,98\ kN/m$$

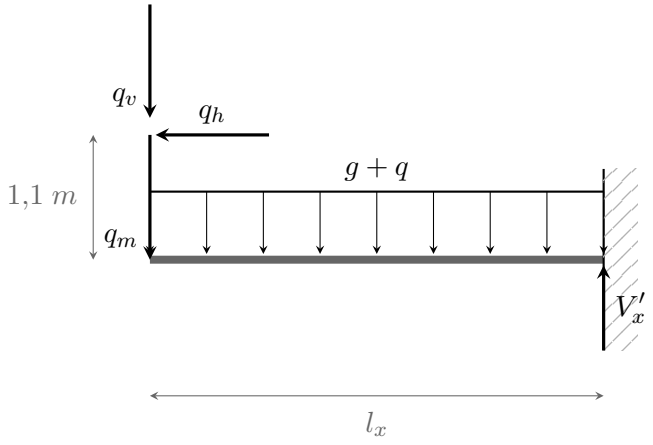


Figura 2 – Corte AA.

$g_m = 1\ kN/m$ Valor estimado da mureta (ABNT NBR 6120, 2019, item 6.3)

$q_v = 2\ kN/m$ carga acidental (ABNT NBR 6120, 2019, tabela 10, índice j)

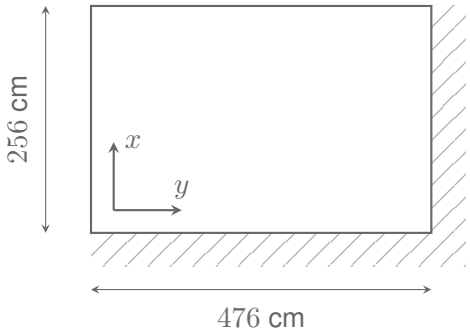
$q_h = 1\ kN/m$ (ABNT NBR 6120, 2019, tabela 12)

$$V'_x = (g+q)\ell_x + g_m + q \rightarrow g_m = 1,1\ h_m$$
$$V'_x \approx 6,75 \times 1,53 + 1,1 \times 1 + 2$$
$$V'_x \approx 13,43\ kN/m$$

L4

b) Método tabelado

L1



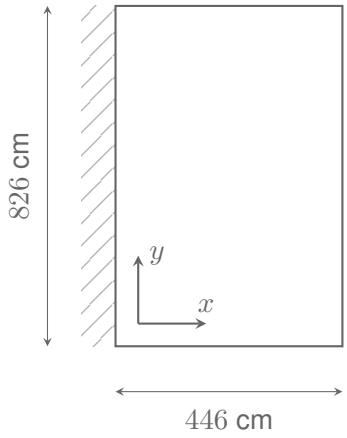
Laje tipo 3 (TABELAS DE LAJES, 2007, tabela 2.2b)

Por interpolação linear:

λ	v_x	v'_x	v_y	v'_y
1,86	3,17	4,64	2,17	2,17

$$V_x = v_x(g+q)\frac{l_x}{10} = 3,17(6,75)\frac{2,56}{10} = 5,47\ kN/m$$
$$V'_x = v'_x(g+q)\frac{l_x}{10} = 8,01\ kN/m$$
$$V_y = v_y(g+q)\frac{l_x}{10} = 3,75\ kN/m$$
$$V'_y = v'_y(g+q)\frac{l_x}{10} = 5,48\ kN/m$$

L2



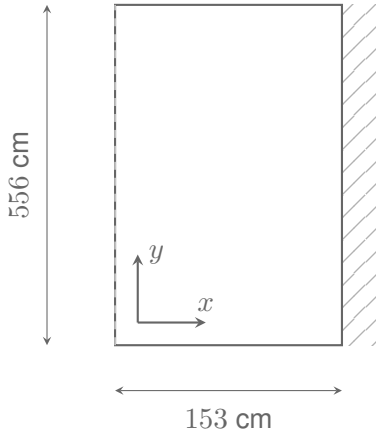
Laje tipo 2B (TABELAS DE LAJES, 2007, tabela 2.2a)

Por interpolação linear:

λ	v_x	v'_x	v_y
1,85	3,47	5,09	1,83

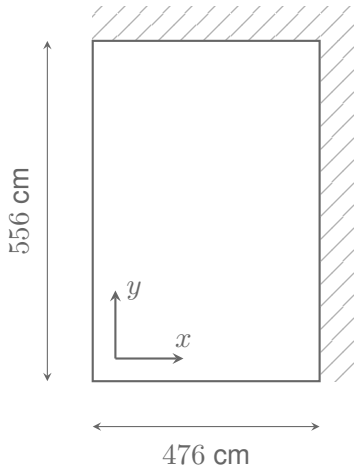
$$V_x = v_x(g+q)\frac{l_x}{10} = 10,45\ kN/m$$
$$V'_x = v'_x(g+q)\frac{l_x}{10} = 15,33\ kN/m$$
$$V_y = v_y(g+q)\frac{l_x}{10} = 5,51\ kN/m$$

L3



Não dá para fazer por tabela, tem que ser por charneiras plásticas.

L4



Laje tipo 3 (TABELAS DE LAJES, 2007, tabela 2.2b)

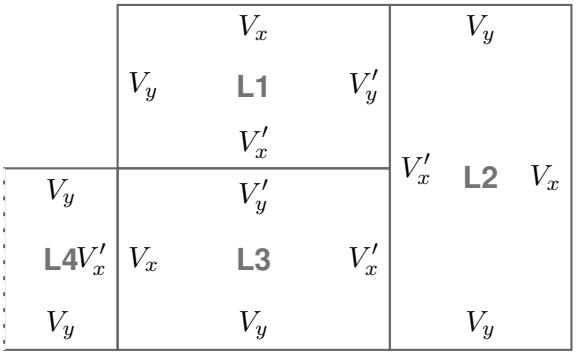
Por interpolação linear:

λ	v_x	v'_x	v_y	v'_y
1,17	2,48	3,62	2,17	3,17

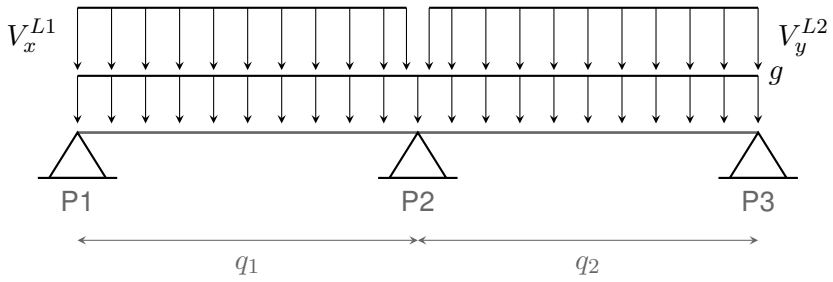
$$V_x = v_x(g+q)\frac{l_x}{10} = 7,96\ kN/m$$
$$V'_x = v'_x(g+q)\frac{l_x}{10} = 11,64\ kN/m$$
$$V_y = v_y(g+q)\frac{l_x}{10} = 6,97\ kN/m$$
$$V'_y = v'_y(g+q)\frac{l_x}{10} = 10,19\ kN/m$$

Aula 03a — 02 de setembro de 2026

5. Determinação do carregamento atuando nas vigas

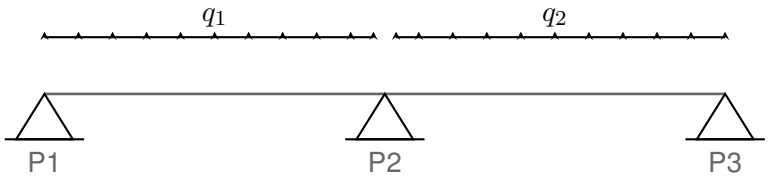


V1



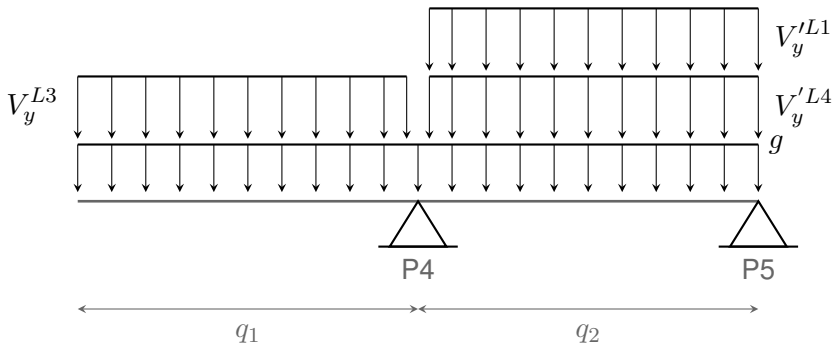
$g = \gamma_{ap} \cdot A_{viga} \rightarrow \gamma_{ap} = 25\ kN/m^3$ (ABNT NBR 6120, 2019, tabela 1)

$$q_1 = g + V_x^{'L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 5,47 = 7,72\ kN/m$$
$$q_2 = g + V_y^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 1,83 = 4,08\ kN/m$$



Continuação - para casa

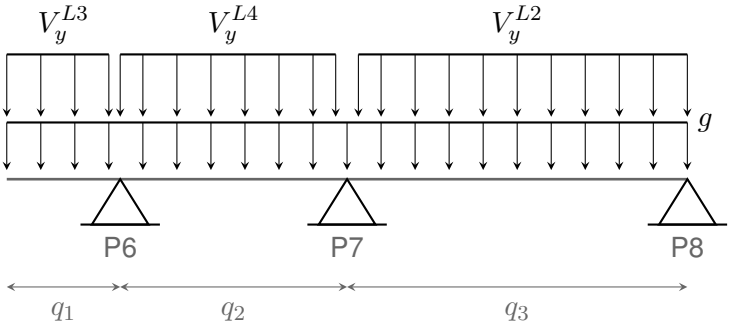
V2



$$q_1 = g + V_x^{\prime L3} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 3 = 5,25\ kN/m$$

$$q_2 = g + V_y^{\prime L4} + V_y^{\prime L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 10,19 + 5,48 = 17,92\ kN/m$$

V3

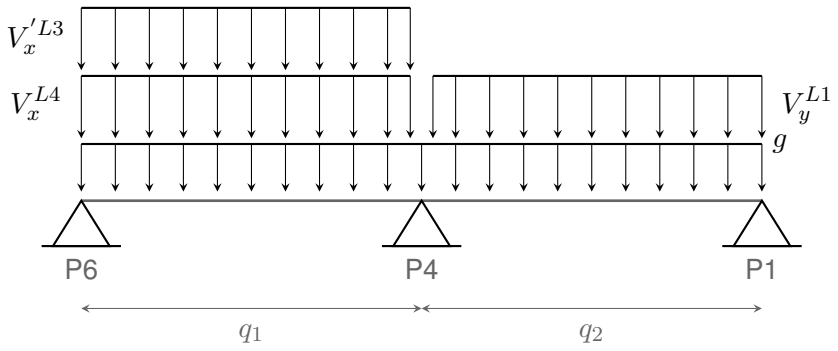


$$q_1 = g + V_x^{\prime L3} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 3 = 5,25\ kN/m$$

$$q_2 = g + V_y^{\prime L4} + V_y^{\prime L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 6,97 = 9,22\ kN/m$$

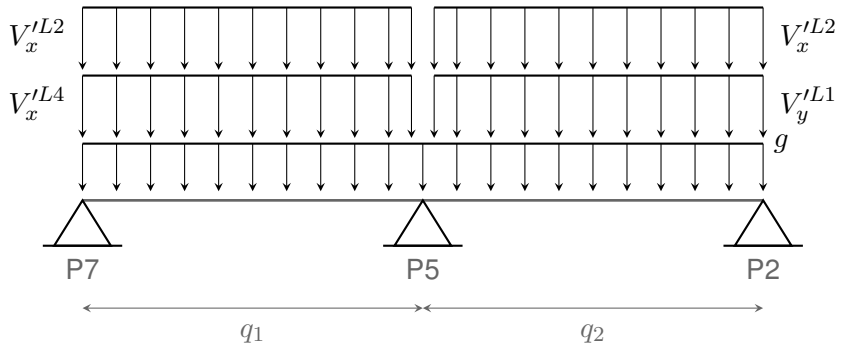
$$q_2 = g + V_y^{\prime L4} + V_y^{\prime L1} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 5,51 = 7,76\ kN/m$$

V4



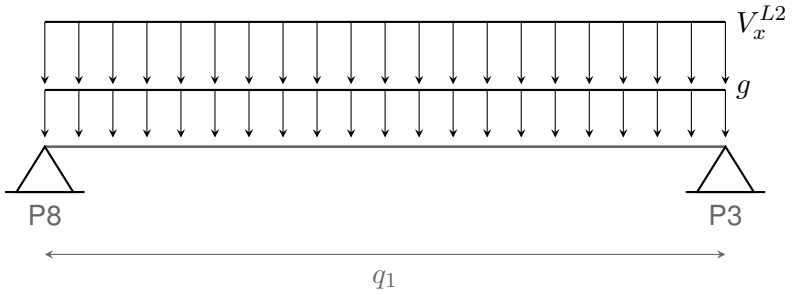
$$q_1 = g + V_x^{'L4} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 11,78 + 7,96 = 21,99\ kN/m$$
$$q_2 = g + V_y^{'L1} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 3,75 = 6\ kN/m$$

V5



$$q_1 = g + V_x^{'L4} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 11,64 + 15,33 = 29,22\ kN/m$$
$$q_2 = g + V_y^{'L1} + V_x^{'L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 5,48 + 15,33 = 23,06\ kN/m$$

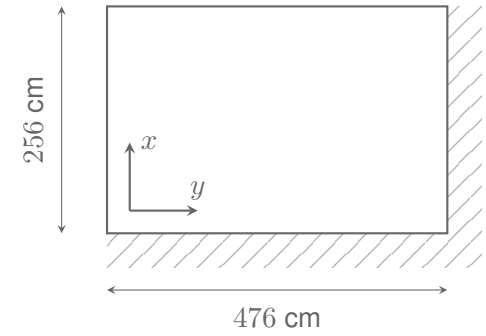
V6

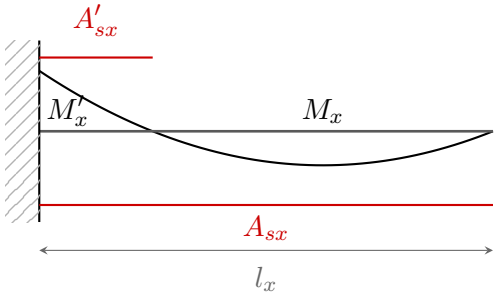


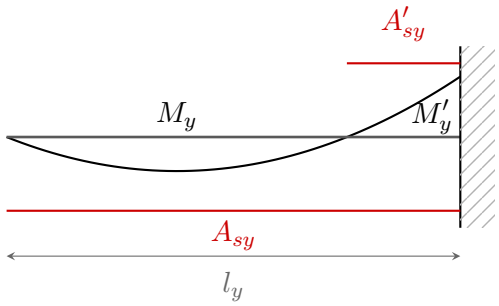
$$q_1 = g + V_x^{L2} = (20 \cdot 10^{-2} \times 45 \cdot 10^{-2})\ 25 + 10,45 = 14,2\ kN/m$$

6. Esforços solicitantes - momentos fletores

L1







Laje tipo 3 (TABELAS DE LAJES, 2007, tabela 2.3b)

λ	μ_x	μ'_x	μ_y	μ'_y
1,86	5,50	11,57	1,71	8,17

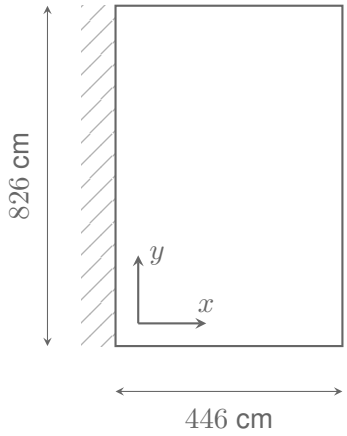
$$M_x = \mu_x(g+q)\frac{\ell_x^2}{100} = 5,50 \times 6,75\ \frac{2,56^2}{100} = 2,43\ kNm/m$$

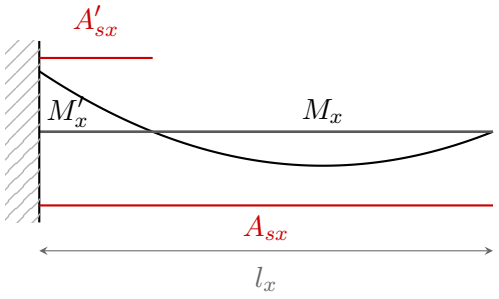
$$M'_x = \mu'_x(g+q)\frac{\ell_x^2}{100} = 5,12\ kNm/m$$

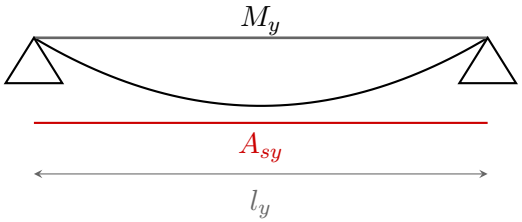
$$M_y = \mu_y(g+q)\frac{\ell_x^2}{100} = 0,76\ kNm/m$$

$$M'_y = \mu'_y(g+q)\frac{\ell_x^2}{100} = 3,61\ kNm/m$$

L2







Laje tipo 2B (TABELAS DE LAJES, 2007, tabela 2.3a)

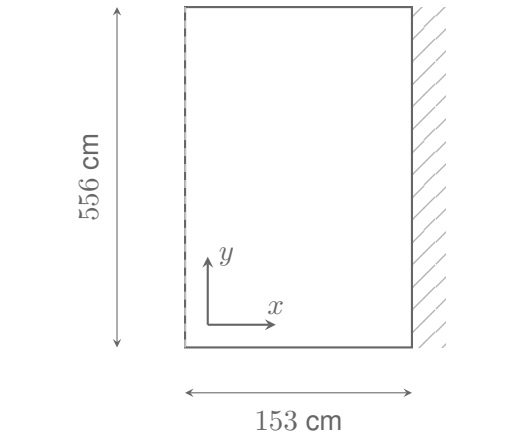
λ	μ_x	μ'_y	μ'_y
1,85	5,81	11,94	1,67

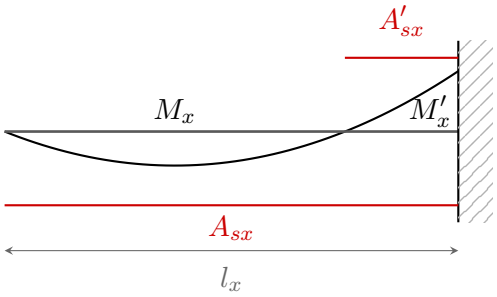
$$M_x = \mu_x(g+q)\frac{\ell_x^2}{100} = 7,80\ kNm/m$$

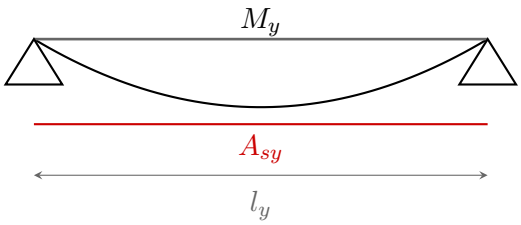
$$M'_x = \mu'_x(g+q)\frac{\ell_x^2}{100} = 16,04\ kNm/m$$

$$M'_y = \mu'_y(g+q)\frac{\ell_x^2}{100} = 2,24\ kNm/m$$

L3







Laje tipo 9 (TABELAS DE LAJES, 2007, tabela 2.3e)

$$\gamma = \frac{\ell_a}{\ell_b} = \frac{1,53}{5,56} = 0,28$$

γ	μ_x	μ'_x	μ_y	μ_{yb}
0,28	-12,50	50	0,78	6,22

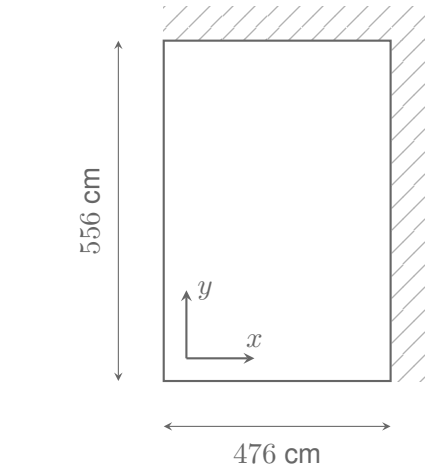
$$M_x = \mu_x \; g \; \frac{\ell_x^2}{100} = -12,50 \times 3,75 \; \frac{1,53^2}{100} = -1,10 \; kNm/m$$

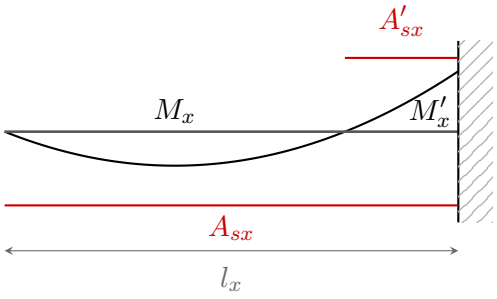
$$M'_x = \mu'_x(g+q)\frac{\ell_a^2}{100} + (g_m + q_v)\; \ell_a + q_h \times L_m = 13,59 \; kNm/m$$

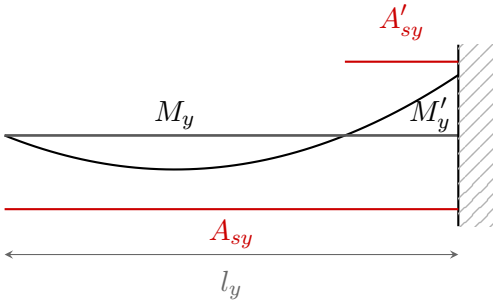
$$M_y = \mu_y \; g\frac{\ell_a^2}{100} = 0,78 \times 3,75 \times \frac{1,53^2}{100} = 0,07 \; kNm/m$$

$$M_{yb} = \mu_{yb}(g+q)\frac{\ell_x^2}{100} = 6,22 \times 6,75 \; \frac{1,53^2}{100} = 0,98 \; kNm/m$$

L4







Laje tipo 3 (TABELAS DE LAJES, 2007, tabela 2.3b)

λ	μ_x	μ'_x	μ_y	μ'_x
1,17	3,51	8,43	2,44	7,55

$$M_x = \mu_x(g+q)\frac{\ell_x}{100} = 5,36\ kNm/m$$

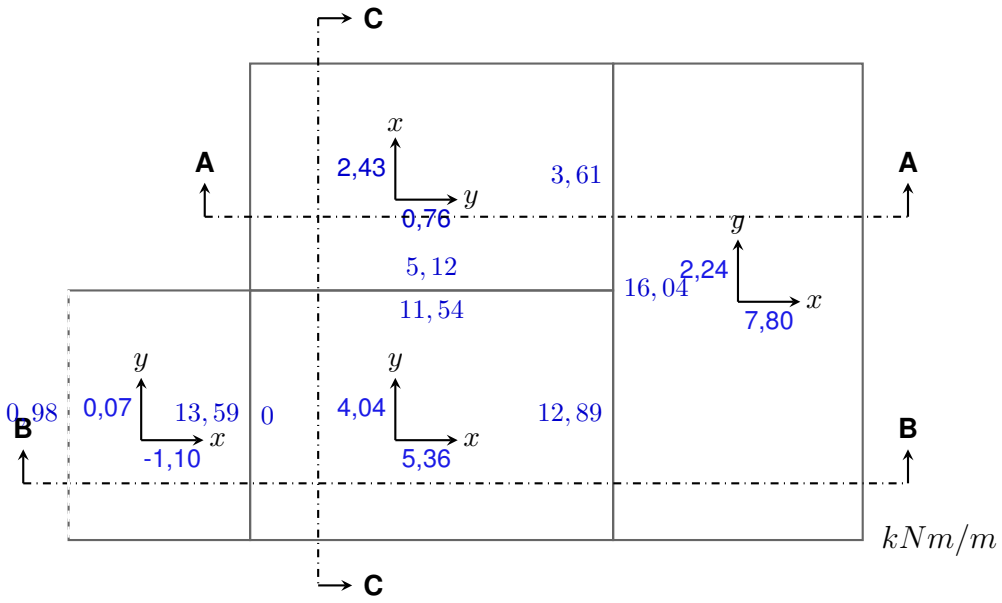
$$M'_x = \mu'_x(g+q)\frac{\ell_x}{100} = 12,89\ kNm/m$$

$$M_y = \mu_y(g+q)\frac{\ell_x}{100} = 4,04\ kNm/m$$

$$M'_y = \mu'_y(g+q)\frac{\ell_x}{100} = 11,54\ kNm/m$$

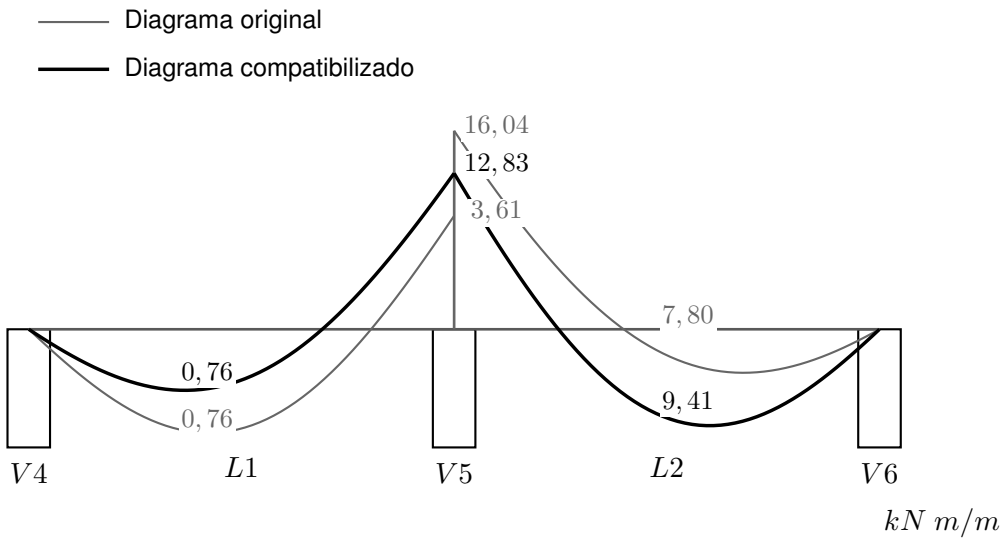
Aula 03b — 03 de setembro de 2026

7. Compatibilização dos momentos fletores



a) Método por cálculo

i. Corte AA

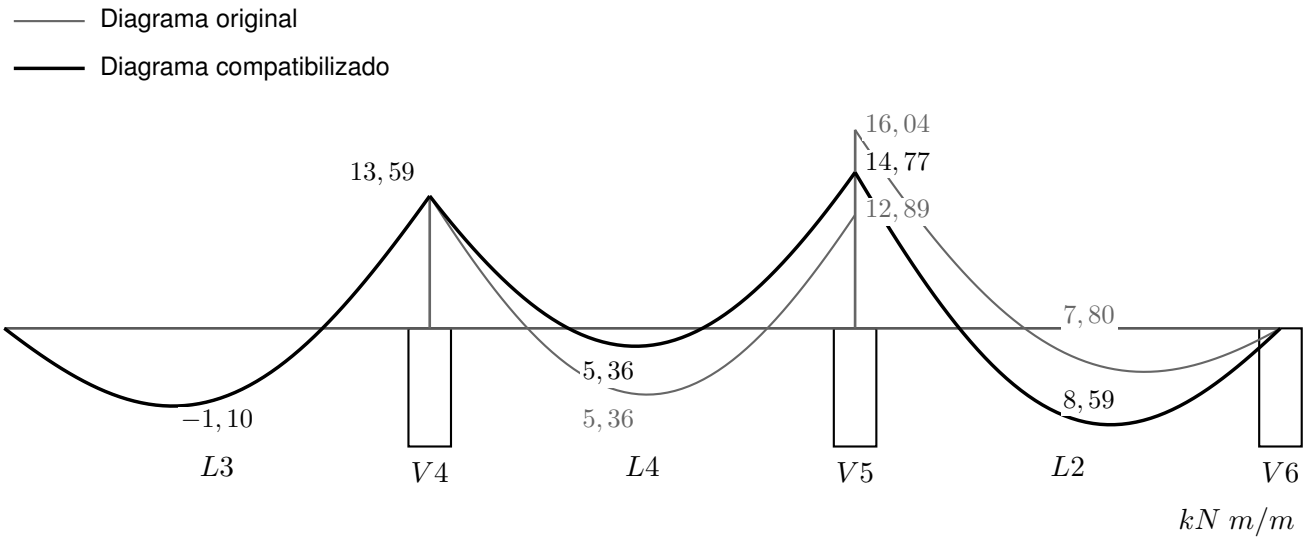


Não precisa fazer a compatibilização da L1 para ficar a favor da segurança.
refendobservacao

$$M_{med} \geq \begin{cases} \dfrac{M' + M'_0}{2} = \dfrac{16,04 + 3,61}{2} = 9,83\ kNm/m \\ 0,8M' = 0,8 \times 16,04 = 12,83\ kNm/m \end{cases}$$

$$M_{c,\ L2} = M_{eixo} + \frac{M' - M_{med}}{2} = 7,8 + \frac{16,04 - 12,83}{2} = 9,41\ kNm/m$$

ii. Corte BB

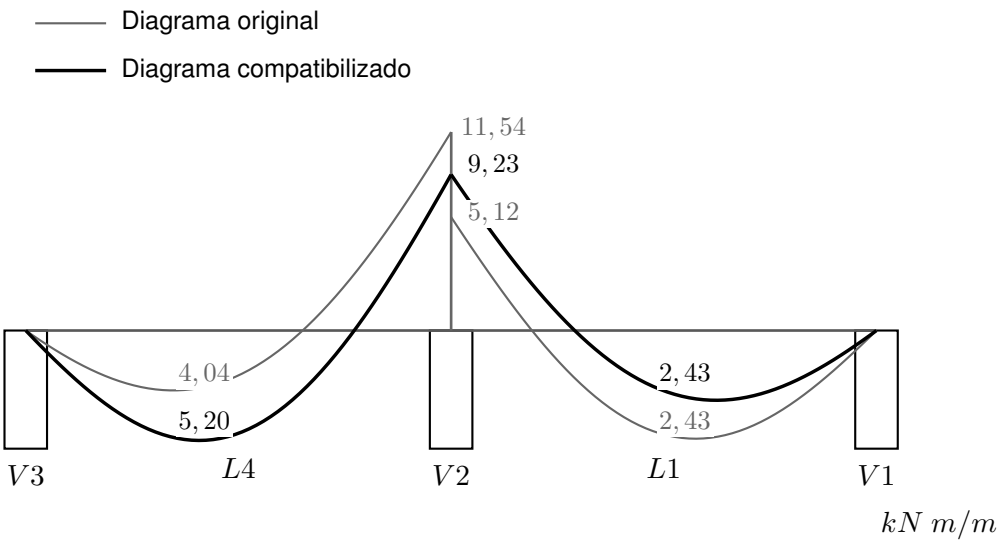


Como a L3 é em balanço, os valores se mantém.

$$M_{med} \geq \begin{cases} \dfrac{M' + M'_0}{2} = \dfrac{16,04 + 12,89}{2} = 14,77\ kNm/m \\ 0,8M' = 0,8 \times 16,04 = 12,83\ kNm/m \end{cases}$$

$$M_{c,\ L2} = M_{eixo} + \frac{M' - M_{med}}{2} = 7,8 + \frac{16,04 - 14,77}{2} = 8,59\ kNm/m$$

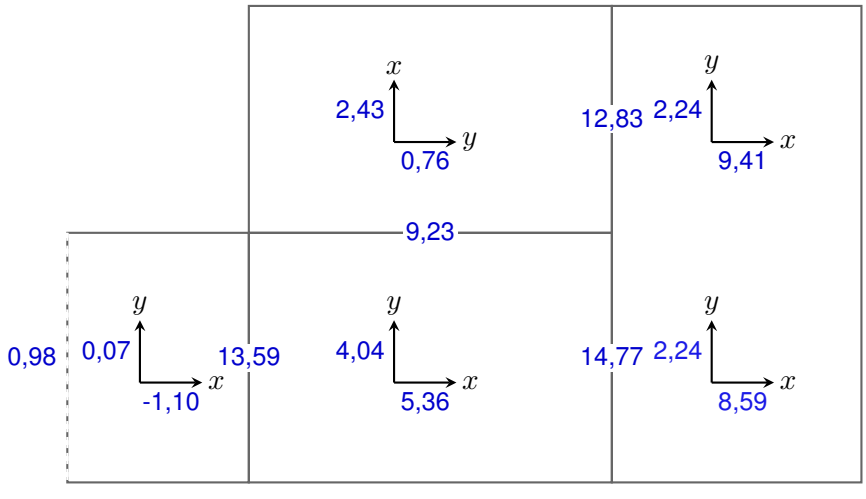
iii. Corte CC



Não precisa fazer a compatibilização da L1 para ficar a favor da segurança.

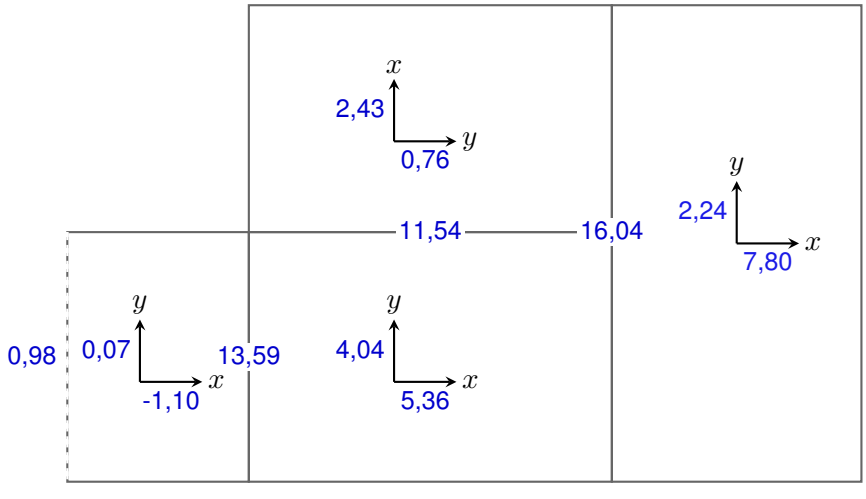
$$M_{med} \geq \begin{cases} \dfrac{M' + M'_0}{2} = \dfrac{11,54 + 5,12}{2} = 8,83\ kNm/m \\ 0,8M' = 0,8 \times 11,54 = 9,23\ kNm/m \end{cases}$$

$$M_{c,\ L4} = M_{eixo} + \frac{M' - M_{med}}{2} = 4,04 + \frac{11,54 - 9,23}{2} = 5,20\ kNm/m$$



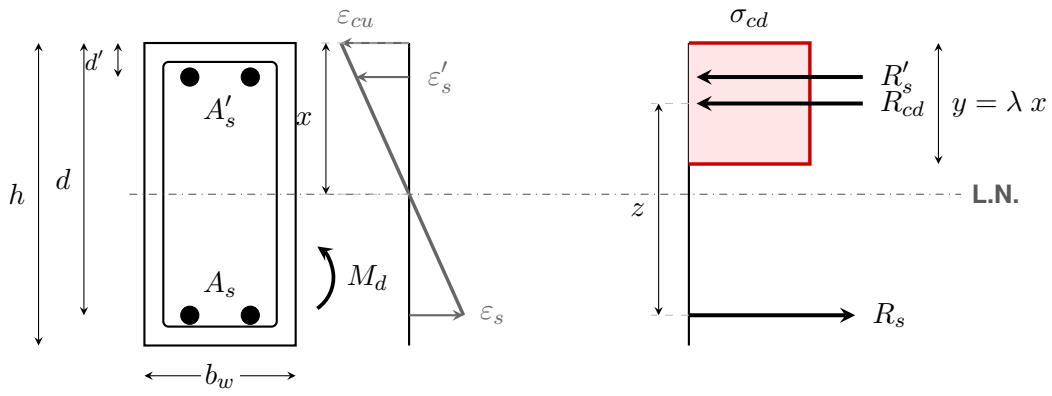
kNm/m

b) Método ABNT



kNm/m

Dimensionamento a flexão por tabelas Kc e Ks



Equilíbrio de momento

$$M_d = R_c\left(d - \frac{y}{2}\right) + R's(d - d'))$$

$$M_d = (b_w\ y\ \sigma_{cd})\left(d - \frac{y}{2}\right) + (A's\ \sigma'_s)(d - d')$$

$$M_d = (b_w\ \lambda\ x\ \alpha_c\ \eta_c\ f_{cd})\left(d - \frac{\lambda\ x}{2}\right) + (A's\ \sigma'_s)(d - d') \rightarrow \beta_x = \frac{x}{d} \leftrightarrow x = \beta_x\ d$$

$$M_d = b_w\ \lambda\ \beta_x\ d\ \alpha_c\ \eta_c\ f_{cd}\left(d - \frac{\lambda\ \beta_x\ d}{2}\right) + \cancelto{0}{(A's\ \sigma'_s)(d - d')}$$

$$M_d = b_w\ \lambda\ \beta_x\ d^2\ \alpha_c\ \eta_c\ f_{cd}\left(1 - \frac{\lambda\beta_x}{2}\right)$$

$$\rightarrow f_{ck} = 40\ MPa\ (\lambda = 0,8, \eta_c = 1, \alpha_c = 0,85)$$

$$M_d = b_w\ \lambda\ \beta_x\ d^2\ \alpha_c\ \eta_c\ f_{cd}\left(1 - \frac{\lambda\beta_x}{2}\right)$$

$$M_d = 0,68\ b_w\ d^2\ f_{cd}\ \beta_x(1 - 0,4\beta_x) \longleftrightarrow \frac{b_w\ \cancel{d^2}}{M_d} = \frac{1}{0,68\ \beta_x\ f_{cd}(1 - 0,4\beta_x)} \quad K_c = f(\beta_x, f_{cd})$$

Do equilíbrio de tensões

$$M_d = R_s\left(d - \frac{y}{2}\right)$$

$$M_d = A_s\ \sigma_{sd}\left(d - \frac{\lambda\ x}{2}\right) \rightarrow \beta_x = \frac{x}{d} \leftrightarrow x = \beta_x\ d$$

$$M_d = A_s\ \sigma_{sd}\left(d - \frac{\lambda\ \beta_x\ d}{2}\right)$$

$$\rightarrow f_{ck} < 50\ MPa(\lambda = 0,80)$$

$$M_d = A_s\ \sigma_{sd}\left(d - \frac{\lambda\ \beta_x\ d}{2}\right)$$

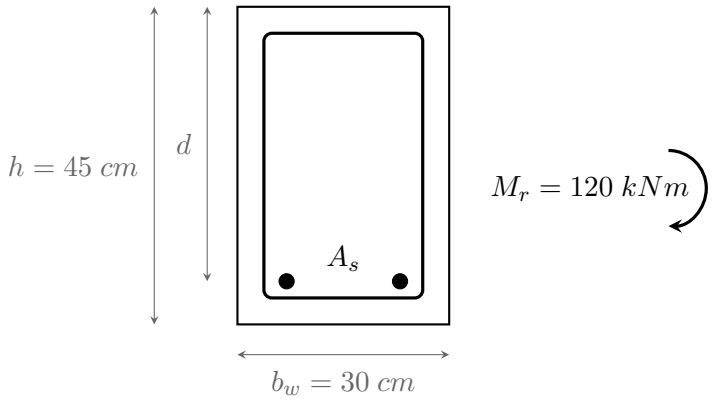
$$M_d = A_s\ \sigma_{sd}\ d\left(1 - 0,4\ \beta_x\right)$$

$$K_s = f(\sigma_s, \beta_x)$$

$$\frac{A_s\ \sigma_{sd}}{M_d} = \frac{1}{1 - 0,4\ \beta_x}$$

Exemplo

1. Calcule a área de aço. Dados: C25, aço CA-50, CAA-II.



a) Momento de projeto

$\gamma_c = 1,4$ (ABNT NBR 6118, 2023, tabela 12.1)

$$M_d = \gamma_c\ M_r = 1,4 \times 120 = 168\ kNm = 16\,800\ kN\ cm$$

b) Altura útil

$$d = h - c - \phi_e - n\frac{\phi_\ell}{2} = 45 \cdot 10^{-2} - 25 \cdot 10^{-3} - 5 \cdot 10^{-3} - 2\ \frac{10 \cdot 10^{-3}}{2} = 41\ cm$$

c) Kc

$$K_c = \frac{b_w\ d^2}{M_d} = \frac{30 \times 41^2}{16800} = 3\ cm^2/kN$$

d) Tipo de armadura

$\beta_x = 0,32 \leq 0,45 \rightarrow$ armadura simples, domínio 3 (SILVA et al., 2023, tabela 1.1a)

e) Ks

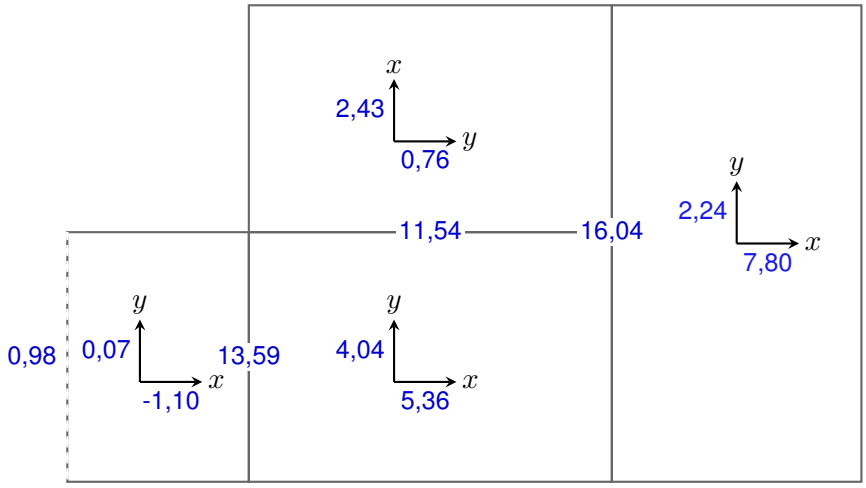
$K_s = 0,026$ (SILVA et al., 2023, tabela 1.1a)

f) Área de aço

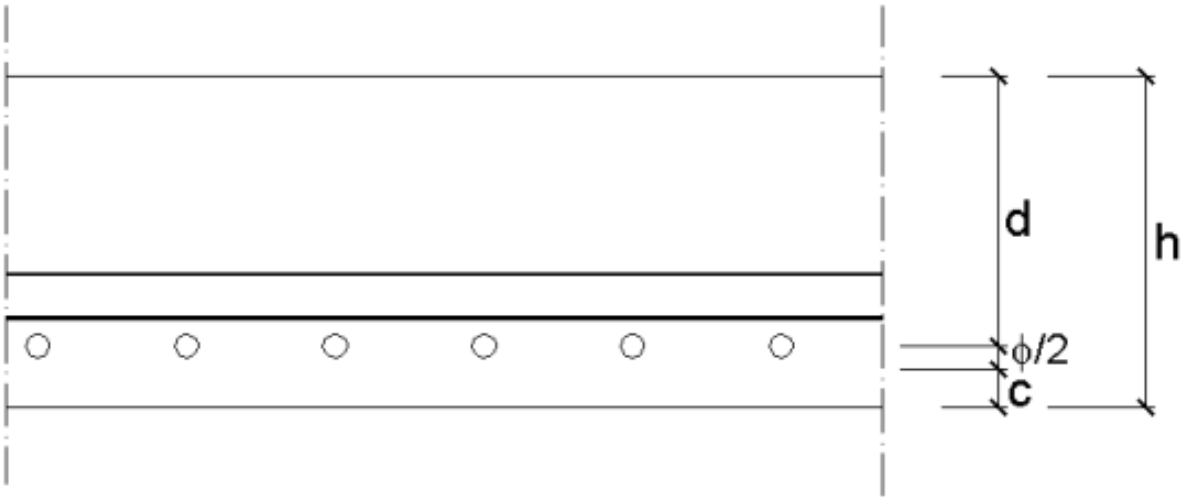
$$K_s = \frac{A_s\ d}{M_d} \longleftrightarrow A_s = \frac{K_s\ M_d}{d} = \frac{0,026 \times 16800}{41} = 10,65\ cm^2$$

Aula 04a — 09 de setembro de 2026

8. Dimensionamento armadura positiva



kNm/m



L1

a) Momento de projeto

$\gamma_c = 1,4$ (ABNT NBR 6118, 2023, tabela 12.1)

$$M_{d,x} = \gamma_c M_x = 1,4 \times 2,43 = 340,2\ kNcm/m$$
$$M_{d,y} = \gamma_c M_y = 1,4 \times 0,76 = 106,4\ kNcm/m$$

b) Altura útil

CAA-I = 20 mm (laje) (ABNT NBR 6118, 2023, tabela 7.2)

$$d = h - c - \phi_e - \frac{\phi_\ell}{2} = 10 \cdot 10^{-2} + \left(-20 - 10 - \frac{10}{2} \right) \cdot 10^{-3} = 6,5\ cm$$

c) Taxa mínima de armadura

$\rho_s = \dfrac{A_s}{A_c} \rightarrow A_c = bh$ (ABNT NBR 6118, 2023, tabela 19.1)

$\rho_{min} = 0,179\%$ (ABNT NBR 6118, 2023, tabela 17.3)

$\rho_s \geq 0,67\rho_{min}$(ABNT NBR 6118, 2023, tabela 19.1)

$A_{s,min} \geq 0,67 \dfrac{0,179 \times 100 \times 10}{100}$

$A_{s,min} \geq 1,2\ cm^2$

d) Área de aço

$K_{c,x} = \dfrac{b\ d^2}{M_d} = \dfrac{100 \times 6,5^2}{340,2} = 12,42\ cm^2/kN$

$K_{c,y} = \dfrac{b\ d^2}{M_d} = \dfrac{100 \times 6,5^2}{106,4} = 39,71\ cm^2/kN$

$\beta_{x,x} = 0,06 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$\beta_{x,y} = 0,02 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$\begin{cases} A_s \geq A_{s,min} \rightarrow A_s \\ A_s < A_{s,min} \rightarrow A_{s,min} \end{cases}$

$K_{s,x} = \dfrac{A_s\ d}{M_d} \longleftrightarrow A_s = \dfrac{K_s\ Md}{d} = \dfrac{0,024 \times 340,2}{6,5} = 1,26\ cm^2/m$

$K_{s,y} = \dfrac{A_s\ d}{M_d} \longleftrightarrow A_s = \dfrac{K_s\ Md}{d} = \dfrac{0,023 \times 106,4}{6,5} = 0,38\ cm^2/m$

e) Determinação da bitola e do espaçamento

$\phi_\ell \leq \dfrac{h}{8} = \dfrac{10}{8} = 1,25\ cm = 12,5\ mm$ (ABNT NBR 6118, 2023, item 20.1)

(ABNT NBR 6118, 2023, item 20.1)

$S_x \leq \begin{cases} 2h = 2 \times 10 = 20\ cm \\ 20\ cm \end{cases} \qquad S_y \leq 33\ cm$

$A_{\phi,x} = \pi\left(\dfrac{D}{2}\right)^2 = \pi\left(\dfrac{5 \cdot 10^{-3}}{2}\right)^2 = 0,196\ cm^2$

O espaçamento deve ser arredondado para baixo.

$S_{\phi,x} = \dfrac{A_\phi}{A_s}\ 100 = \dfrac{0,196}{1,26}\ 100 = 15,58 \approx 15\ cm$

$\phi_{\ell,x} = 5\ mm$ c/ $15\ cm$

$$S_{\phi,y} = \frac{A_\phi}{A_s}\,100 = \frac{0,196}{1,2}\,100 = 16,36 \approx 16\ cm$$
$$\phi_{\ell,y} = 5\ mm \text{ c/ } 16\ cm$$

L2

a) Momento de projeto

$$M_{d,x} = \gamma_c M_x = 1,4 \times 7,80 = 1092\ kNcm/m$$
$$M_{d,y} = \gamma_c M_y = 1,4 \times 2,24 = 313,6\ kNcm/m$$

b) Área de aço

$$K_{c,x} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{1092} = 3,87\ cm^2/kN$$
$$K_{c,y} = \frac{b\,d^2}{M_d} = \frac{100 \times 6,5^2}{313,6} = 13,47\ cm^2/kN$$

$\beta_{x,x} = 0,14 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$\beta_{x,y} = 0,02 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$$K_{s,x} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,025 \times 1092}{6,5} = 4,20\ cm^2/m$$
$$K_{s,y} = \frac{A_s\,d}{M_d} \longleftrightarrow A_s = \frac{K_s\,Md}{d} = \frac{0,023 \times 313,6}{6,5} = 1,11\ cm^2/m$$

c) Determinação da bitola e do espaçamento

$$A_{\phi,x} = \pi\left(\frac{D}{2}\right)^2 = \pi\left(\frac{8 \cdot 10^{-3}}{2}\right)^2 = 0,5\ cm^2$$
$$S_{\phi,x} = \frac{A_\phi}{A_s}\,100 = \frac{0,5}{4,2}\,100 = 11,98 \approx 11\ cm$$
$$\phi_{\ell,x} = 8\ mm \text{ c/ } 11\ cm$$

$$S_{\phi,y} = \frac{A_\phi}{A_s}\,100 = \frac{0,196}{1,2}\,100 = 16,36 \approx 16\ cm$$
$$\phi_{\ell,y} = 5\ mm \text{ c/ } 16\ cm$$

L3

Corrigir

L4

a) Momento de projeto

$M_{d,x} = \gamma_c M_x = 1,4 \times 5,36 = 750,4\ kNcm/m$

$M_{d,y} = \gamma_c M_y = 1,4 \times 4,04 = 565,6\ kNcm/m$

b) Área de aço

$$K_{c,x} = \frac{b\ d^2}{M_d} = \frac{100 \times 6,5^2}{750,4} = 5,63\ cm^2/kN$$

$$K_{c,y} = \frac{b\ d^2}{M_d} = \frac{100 \times 6,5^2}{565,6} = 7,47\ cm^2/kN$$

$\beta_{x,x} = 0,10 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$\beta_{x,y} = 0,08 \leq 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$$K_{s,x} = \frac{A_s\ d}{M_d} \longleftrightarrow A_s = \frac{K_s\ Md}{d} = \frac{0,024 \times 750,4}{6,5} = 2,77\ cm^2/m$$

$$K_{s,y} = \frac{A_s\ d}{M_d} \longleftrightarrow A_s = \frac{K_s\ Md}{d} = \frac{0,024 \times 565,6}{6,5} = 2,09\ cm^2/m$$

c) Determinação da bitola e do espaçamento

$$A_{\phi,x} = \pi\left(\frac{D}{2}\right)^2 = \pi\left(\frac{8 \cdot 10^{-3}}{2}\right)^2 = 0,5\ cm^2$$

$$S_{\phi,x} = \frac{A_\phi}{A_s}\ 100 = \frac{0,5}{2,77}\ 100 = 18,05 \approx 18\ cm$$

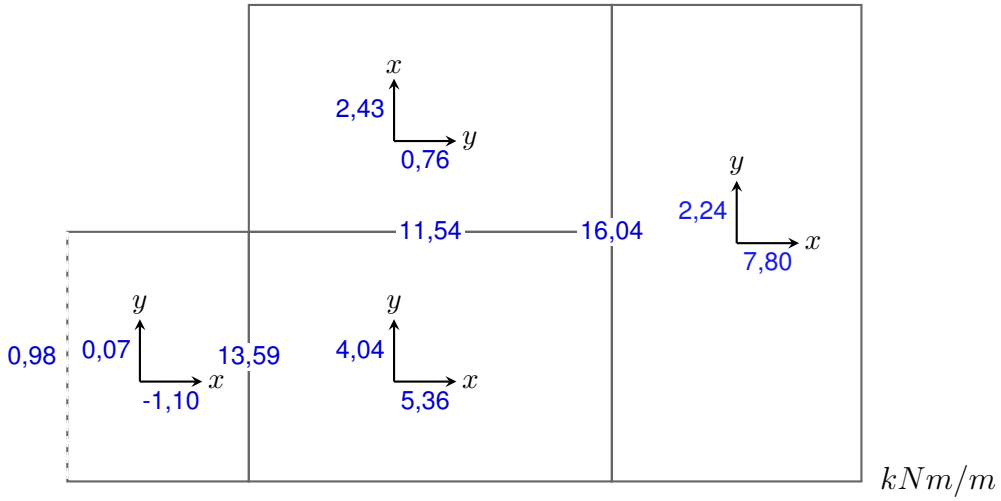
$\phi_{\ell,x} = 8\ mm$ c/ $18,05\ cm$

$$S_{\phi,y} = \frac{A_\phi}{A_s}\ 100 = \frac{0,5}{2,09}\ 100 = 23,92 \approx 23\ cm$$

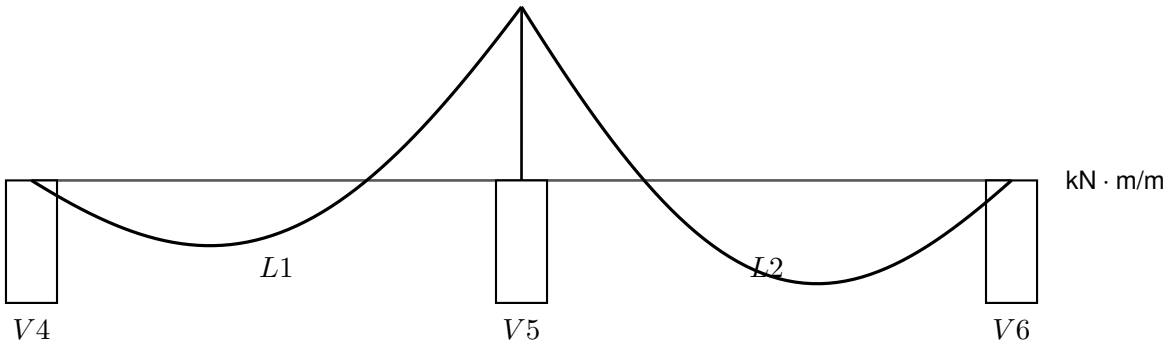
$\phi_{\ell,y} = 8\ mm$ c/ $23\ cm$

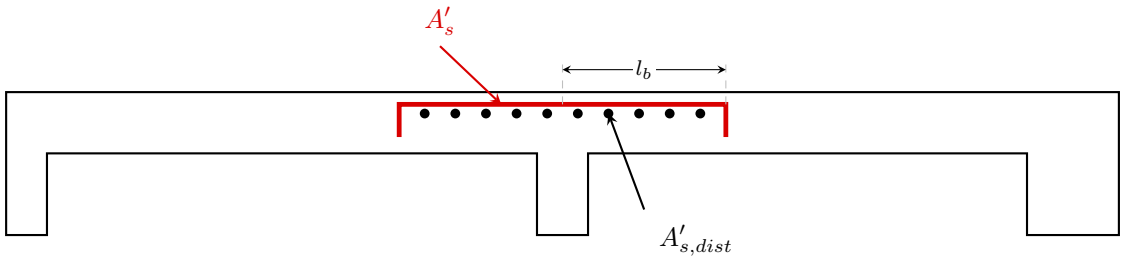
Aula 04b — 10 de setembro de 2026

9. Dimensionamento armaduras negativas



L1-L2 L2-L4





a) Momento de projeto

$M'k = 16,04\ kNm/m$

$\gamma_c = 1,4$ (ABNT NBR 6118, 2023, tabela 12.1)

$M'_d = \gamma_c\ M'k = 1,4 \times 16,04 = 2245,6\ kNcm/m$

b) Altura útil

$$d' = h - c - \frac{\phi_\ell}{2} = 10 \cdot 10^{-2} - 20 \cdot 10^{-3} - \frac{10 \cdot 10^{-3}}{2} = 7,5\ cm$$

c) Taxa mínima de armadura

$\rho_s \geq \rho_{min}$(ABNT NBR 6118, 2023, tabela 17.3 e tabela 19.1)

$$\frac{A_s}{b_w\ h} \geq \frac{0,179}{100} \longleftrightarrow A_s min = \frac{0,179 \times 100 \times 10}{100} = 1,79\ cm^2/m$$

d) Área de aço

$$K_c = \frac{b\ d'^2}{M_d} = \frac{100 \times 7,5^2}{2245,6} = 2,50\ cm^2/kN$$

$\beta_x = 0,24 < 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$$K_s = \frac{A_s\ d'}{M_d} \longleftrightarrow A_s = \frac{0,025 \times 2245,6}{7,5} = 7,49\ cm^2/m$$

e) Armadura de distribuição negativa (ABNT NBR 6118, 2023, item 20.1)

A armadura secundária ou armadura de distribuição serve, apenas, como elementos construtivo.

$$A'_{s,dist} \geq \begin{cases} 20\% A_s = \dfrac{20}{100}\ 7,49 = 1,50\ cm^2 \\ 0,9\ cm^2/m \end{cases}$$

f) Determinação da bitola e do espaçamento

Sempre considerar a altura da menor laje.

$$\phi_{\ell,max} = \frac{h}{8} = 1,25\ cm = 12,5\ mm$$

$$S_p \leq \begin{cases} 2h = 2 \times 10 = 20\ cm \\ 20\ cm \end{cases} \qquad S_d \leq 33\ cm$$

$$A_\phi = \pi\left(\frac{D}{2}\right)^2 \qquad A_\phi = \pi\left(\frac{D}{2}\right)^2$$

$$A_\phi = \pi\left(\frac{12,5 \cdot 10^{-3}}{2}\right)^2 = 1,23\ cm^2 \qquad A_\phi = \pi\left(\frac{6,3 \cdot 10^{-3}}{2}\right)^2 = 0,31\ cm^2$$

$$S_{\phi,p} = \frac{A_\phi}{A_s}\ 100 \qquad S_{\phi,d} = \frac{A_\phi}{A_s}\ 100$$

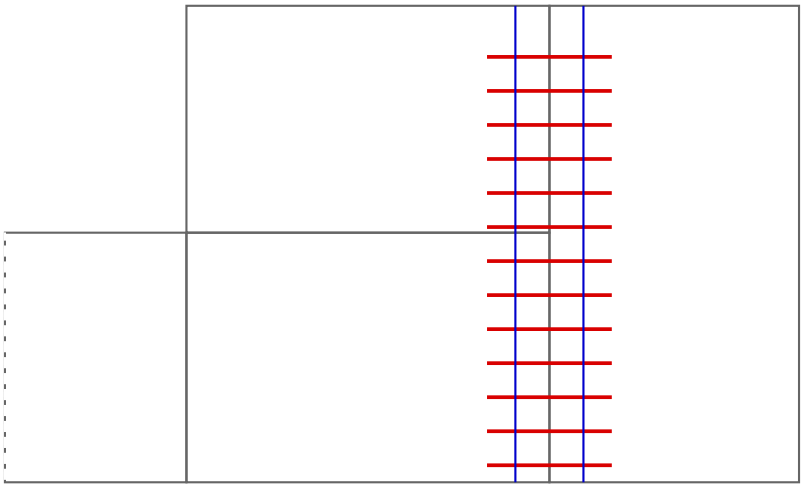
$$S_{\phi,d} = \frac{1,23}{7,49}\ 100 = 16,42 \approx 16\ cm \qquad S_{\phi,d} = \frac{0,31}{1.79}\ 100 = 17,32 \approx 17\ cm$$

$\phi_\ell = 12,5mm$ c/ $16\ cm$ $\qquad$ $\phi_\ell = 6,3mm$ c/ $17\ cm$

ℓ_x vai ser o maior vão efetivo das lajes contínuas.

$$\ell_x \geq \begin{cases} \ell_{L1,L2} \geq \begin{cases} \ell_{x,L1} = 256\ cm \\ \ell_{x,L2} = 446\ cm \end{cases} \\ \ell_{L2,L4} \geq \begin{cases} \ell_{x,L2} = 446\ cm \\ \ell_{x,L4} = 476\ cm \end{cases} \end{cases}$$

$$a_\ell = 0,25\ \ell_x = 0,25 \times 476 = 119\ cm$$



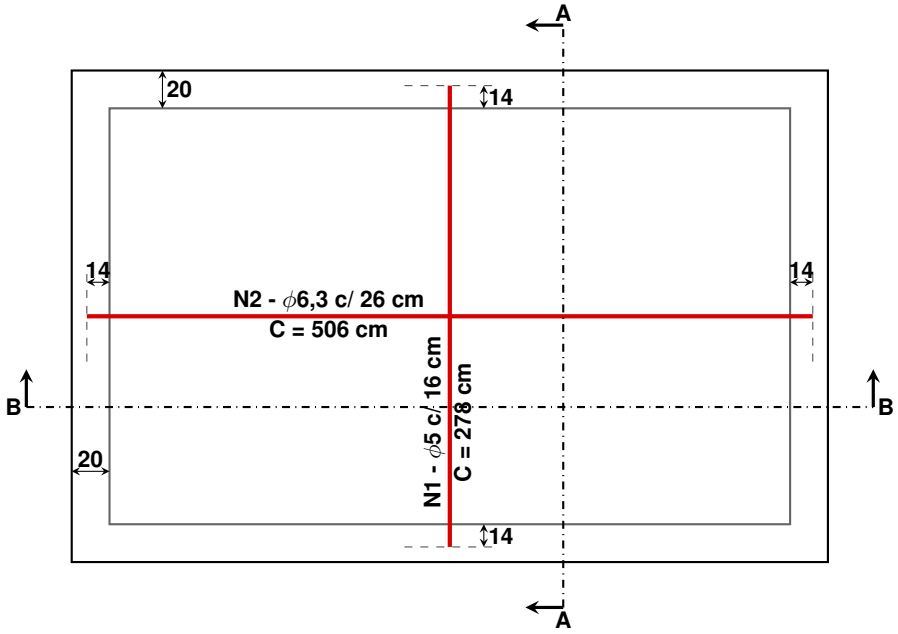
kNm/m

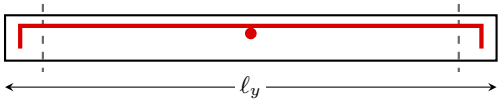
L1-L4

L3-L4

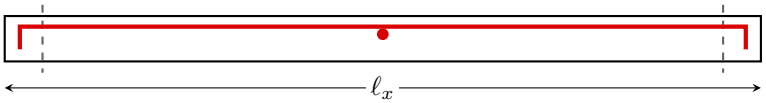
10. Detalhamento das armaduras

L1





CORTE AA



CORTE BB

a) Comprimento de ancoragem básico (ABNT NBR 6118, 2023, item 9.4.2.4)

i. Resistência da aderência de cálculo (ABNT NBR 6118, 2023, item 9.3.2)

$f_{ctk,inf} = 0,7\ f_{ct,m}$ (ABNT NBR 6118, 2023, item 8.2.5)

$f_{ct,m} = 0,3 f_{ck}^{2/3}$ (ABNT NBR 6118, 2023, item 8.2.5)

$\gamma_c = 1,4$ (ABNT NBR 6118, 2023, tabela 12.1)

$$f_{ctd} = \frac{f_{ctk,inf}}{\gamma_c} \text{(ABNT NBR 6118, 2023, tabela 8.2)}$$

$$f_{ctd} = \frac{0,7 \times 0,3 \times 25^{2/3}}{1,4} = 1,28\ MPa$$

$\eta_1 = 2,25$ (ABNT NBR 6118, 2023, tabela 8.2)

$\eta_2 = 1$ (ABNT NBR 6118, 2023, item 9.3.1)

$\eta_3 = 1$ (ABNT NBR 6118, 2023, item 9.3.2.1)

$$f_{bd} = \eta_1\ \eta_2\ \eta_3 f_{ctd} = 2,25 \times 1 \times 1 \times 1,28 = 2,89\ MPa$$

$\gamma_s = 1,15$ (ABNT NBR 6118, 2023, tabela 12.1)

$$f_{yd} = \frac{f_{yk}}{\gamma_s}$$

$$\ell_b = \frac{\phi}{4}\frac{f_{yd}}{f_{bd}} \geq 25\phi$$

$$\ell_b = \frac{\times 0,5}{4}\frac{\frac{500}{1,15}}{2,89} \geq 25 \times 0,5$$

$$\ell_b = 18,81 \geq 12,5$$

b) Comprimento de ancoragem necessário (ABNT NBR 6118, 2023, item 9.4.2.5)

1 para ficar a favor da segurança.

$\alpha = 1$ (ABNT NBR 6118, 2023, item 9.4.2.5)

$$\ell_{b,nec} = \alpha\ \ell_b \frac{A_{s,calc}}{A_{s,ef}} \geq \ell_{b,min}$$

$$\ell_{b,nec} = \alpha\ \ell_b \frac{A_{s,calc}}{A_{s,ef}} \geq \begin{cases} 1 \\ 0,3\ell_b = 0,3 \times 18,81 = 5,64\ cm \\ 10\phi = 10 \times 0,5 = 5\ cm \\ 10\ cm \end{cases}$$

$$\ell_{b,nec} = 1 \times 18,81 = 18,81 \geq 10\ cm$$

L2

L3

L4

Referências

Associação Brasileira de Normas Técnicas. *NBR 6120: Ações para o cálculo de estruturas de edificações*. Rio de Janeiro, 2019. 61 p.

Associação Brasileira de Normas Técnicas. *NBR 6118: Projeto de estruturas de concreto*. Rio de Janeiro, 2023. 242 p.

SILVA, G. d. S. et al. *Tabelas Kc e Ks para fck < 90 MPa*. 1. ed. Barra do Garças: Universidade Federal de Mato Grosso, 2023.

UNIVERSIDADE DE SAO PAULO. *Tabelas de lajes*. Sao Carlos, 2007. 24 p.